

Strategic Impact Mapping: A Directed-Network Framework for S&P 500 Crisis Regimes

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Abstract—Financial crises rewire the topology of asset co-movement: correlations spike, sector blocks dissolve, and a handful of nodes absorb an outsized share of the spillover. We ask whether a single, reproducible pipeline can map that rewiring across four decades of the S&P 500, and measure under strict multiple-testing discipline how much of it withstands scrutiny. The pipeline threads A-DCC GARCH conditional correlations, a Graphical LASSO partial-correlation filter, an edge-level lead/follower cascade, four network-topology metrics, and a composite Network Stress Index (NSI) into one panel-scale workflow, run per snapshot over twenty crisis, stress, recovery, and calm regimes on a survivorship-free CRSP panel. Two elements are new: the end-to-end integration, and a constrained-BIC fallback that recovers a usable graph at the windows where the standard selector collapses to the empty graph. The findings are stated in full. PageRank leadership concentration is the sharpest crisis signature by a clear margin; yet once the whole battery of channels meets a multiplicity correction none survives, and the topology-only NSI ranks the 2008 global financial crisis first but does not separate crisis from calm on average. Exact permutation tests, cluster- and block-bootstrap confidence sets, and an automated 836-check numerical-invariant suite stand behind every claim, the negative results included. We release the framework, its inference layer, and a full companion appendix as a rigorously characterised foundation rather than a finished predictor.

Index Terms—Financial networks, Dynamic conditional correlation, Graphical lasso, Lead-lag structure, Motif analysis, Network stress index, S&P 500.

I. INTRODUCTION

FINANCIAL crises reshape the topology of asset co-movement: pairwise correlations rise, sectoral block structure weakens, and a small number of nodes carry a disproportionate share of the systemic spillover. A growing literature captures each of these effects separately, but to our knowledge no S&P 500 study integrates all four of (i) time-varying conditional correlations, (ii) partial-correlation filtering of shared-factor confounding, (iii) edge-level lead/follower direction assignment, and (iv) aggregation into a cross-window stress reading inside a single panel-scale workflow.

The three closest prior pipelines each share two ingredients with the present construction but stop short of the fourth. Cerqueti et al. [1] pair DCC with partial correlation on the European market but stop at an undirected network (no Stage-3 direction cascade, no triadic motif census, no composite stress reading). Kim et al. [2] add a lead/follower direction step but on a 30-asset universe at most, and without precision-matrix regularisation at $n/p < 1$ or a panel-scale composite. Basnarkov et al. [3] apply GLASSO precision-matrix filtering to a

foreign-exchange panel, not to S&P 500 equities, and without a directed step or a stress index. This paper integrates all four ingredients (A-DCC GARCH; GLASSO partial correlation with adaptive EBIC; a two-test lead/follower cascade; and aggregation into the composite NSI) and applies them to the S&P 500 over 1985–2024 across twenty snapshots spanning seven crisis peaks (1987 Black Monday, 1997 Asian, 1998 LTCM/Russian default, 2000 dot-com, 2001 9/11, 2008 GFC, 2020 COVID), six non-crisis stress windows, two recoveries, and five calm baselines.

Contributions. (i) An end-to-end pipeline integrating A-DCC GARCH, a Graphical LASSO precision matrix with adaptive EBIC, a two-test lead/follower cascade, four network metrics (clustering deviation, PageRank concentration, Louvain modularity with sector purity, MAN triadic motifs), and a composite Network Stress Index (NSI), applied at S&P 500 panel scale across twenty CRSP snapshots; the integration and the Stage-2 numerical modifications below are the novel elements.

(ii) Two Stage-2 numerical modifications, both required for GLASSO to yield a downstream-usable graph at the nineteen $n/p < 0.30$ windows (every snapshot but the 2024 Contemporary baseline): (a) an n/p -tiered identity-target shrinkage schedule that keeps the sklearn coordinate-descent solver inside its SPD-convergence basin, and (b) a constrained-BIC fallback that selects the BIC-optimal λ among grid points clearing an $n_{\text{edges}} \geq p$ floor when the unconstrained argmin lands at the empty graph (§III-B, eq. (2)).

(iii) An empirical test of four ex-ante hypotheses¹ on the historical-constituent CRSP panel (655–889 alive PERMNOs per snapshot):

- Q2 (PageRank concentration) holds in regime-mean direction on Gini and is the only channel to pass the formal crisis-vs-non-crisis permutation test ($p_{\text{Gini}} = 0.022$); the full-network HHI moves in the same direction but is descriptive.
- Q4 (MR motif decline) holds in regime-mean direction on the mutual-dyad fraction; on $|Z_{\text{MR}}|$ it singles out the COVID peak rather than separating crisis from baseline.
- Q3 (community dissolution) is partially supported on the density-matched cross-sector edge fraction but inconclusive on density-matched modularity at the stringent top-779 budget.

¹Hypotheses Q1–Q4 and the small-world coefficient threshold $\sigma > 4$ of §VI-A were specified in the project design document (available from the author on request) before any inference was run on the historical-constituent panel; the design document is not formally pre-registered with an external registry.

- Q1 (clustering deviation) rejects randomness in every regime without discriminating among them.

(iv) A snapshot- and rolling-NSI-VIX concordance with cluster- and block-bootstrap confidence sets: snapshot Pearson $r = 0.321$ has cluster-bootstrap 95% CI $[-0.289, 0.657]$ including zero (§VII-B); the rolling contemporaneous $r = 0.203$ has block-bootstrap 95% CI $[-0.052, 0.398]$ at $\bar{L} = 12$, also including zero and sign-inconsistent across horizons (§VII-C), so neither establishes a significant contemporaneous or leading signal.

(v) A formal inference layer (§VII, expanded in Appendix §J): exact permutation tests on the crisis-vs-non-crisis contrast across seven channels (PageRank Gini the only channel clearing one-sided $p < 0.05$; none clears Benjamini-Hochberg) and per-axis Bonferroni-20 corrected per-snapshot Z -scores for the Q1 and Q4 axes.

§§II–X cover related work, the five-stage pipeline, data and the twenty snapshots, empirical results, density-controlled robustness, formal inference, the consolidated scorecard, limitations with paired future-work items, and reproducibility metadata; the companion appendix covers what the page budget excludes.

II. BACKGROUND AND RELATED WORK

Each of the four pipeline ingredients has a mature literature; no prior pipeline integrates all four on an equity panel. The A-DCC specification of Cappiello et al. [4] extends Engle’s DCC [5] with an asymmetric term that fires under joint negative shocks, matching the observation that crisis correlations climb harder than calm ones. The small-sample bias of the two-stage DCC estimator is corrected by Aielli’s cDCC variant [6], which we do not adopt because the global pooling step on the always-alive subset already absorbs most of the per-window finite-sample noise.

Partial correlation filtered through a Graphical LASSO [7] yields a less factor-confounded financial network [1], [3] by stripping pairwise inflation from shared macro loadings. Kenett et al. [15] apply partial correlation to S&P 500 returns without precision-matrix regularisation; GLASSO is adopted here for the regularisation control at $n/p < 1$ via λ and the EBIC criterion [16] that an unregularised inversion and the older minimum-spanning-tree [12], planar-maximally-filtered [13], and random-matrix-theory [14] filters (surveyed in [11]) do not provide. Earlier sparse Gaussian graphical routes [8] and tail-dependence findings on equity panels [10] stop short of the directed Stage-3 step.

Directed information flow needs a separate test. Kim et al. [2] combine lagged partial correlation with Granger causality [17], [18], [19]; related panel-scale work (Curme et al. [20], Stavroglou et al. [21]) relies on a single test rather than the two-test cascade here, which is motivated because the Granger test alone fires bidirectionally on up to 8.8% of Stage-2 edges (peak at Mar 2020; §III-C).

The network metrics trace to standard references [22], [23], [24], [25], [28]. DebtRank [29] and the systemic-risk literature [31] require firm-level exposure data unavailable here; spillover routes [30] fit a system rather than edge-level

directions, and CoVaR [32] operates at the institution-pair level. Composite financial-stress indices [33], [34] aggregate balance-sheet/market/credit signals; the Stage-5 NSI of §III-E is a topology-only counterpart at the equity-network level.

Table I locates the present pipeline against the six closest constructions in the literature on the four-axis ingredient checklist (time-varying conditional correlation, partial-correlation filtering, edge-level direction assignment, panel-scale topological composite). The integration of all four axes at S&P 500 panel scale across twenty regime snapshots is what distinguishes this work; no surveyed pipeline carries all four ingredients on an equity panel of this size.

TABLE I – Closest prior pipelines on a four-axis ingredient checklist: (T) time-varying conditional correlation; (P) partial-correlation filtering; (D) edge-level direction assignment; (C) panel-scale topological composite. Universe lists the asset count and market. “Yes*” marks partial implementations (single-test direction, sub-panel size).

Pipeline	T	P	D	C	Universe
Cerqueti et al. [1]	✓	✓			EU equities
Kim et al. [2]	✓	✓	✓*		30 KOSPI tickers
Basnarkov et al. [3]		✓			21 FX pairs
Kenett et al. [15]		✓			S&P 500
Diebold–Yilmaz [30]	✓		✓*	✓	16 G-7 assets
CISS [33]				✓	EU macro indicators
This paper	✓	✓	✓	✓	S&P 500 (CRSP, 655–889 alive)

III. METHODOLOGY

Five stages turn a daily return panel into one stress reading (Fig. 1): A-DCC GARCH conditional correlations, a Graphical LASSO partial-correlation filter, a two-test direction-assignment cascade, four network metrics, and a composite NSI. The Stage-3 directed graph with its four Stage-4 metrics is the *strategic impact map* of a snapshot (three are rendered in Fig. 2 of §V). Decimals are rounded half-up; relative differences use unrounded inputs, so recomputing from displayed values may shift the last digit. Symbols are defined at first use; the full glossary is in Appendix §A.

A. Stage 1: A-DCC GARCH Estimation

Univariate GARCH(1,1) [35] with Student- t innovations strips conditional volatility and heavy tails from each asset; the standardised residuals z_{it} feed the scalar A-DCC recursion of Cappiello, Engle, and Sheppard [4],

$$Q_t = (1 - a - b)\bar{Q} - g\bar{N} + a z_{t-1} z'_{t-1} + b Q_{t-1} + g n_{t-1} n'_{t-1},$$

with element-wise $n_t = \min(z_t, 0)$, sample targets $\bar{Q} = T^{-1} \sum_t z_t z'_t$ and $\bar{N} = T^{-1} \sum_t n_t n'_t$, and the normalisation $R_t = D_t Q_t D_t$ where $D_t = \text{diag}(Q_t)^{-1/2}$.

Globally pooled (a, b, g). Re-estimating per window would mix parameter drift with window length, so the parameters are fit once with multi-start L-BFGS-B on a 100-asset balanced subset of the always-alive set ($\geq 99\%$ coverage; 278 tickers qualify, the first 100 enter the fit [6], [36]). The pre-MLE residual panel is clipped to $|z| \leq 10$ so the three severe crash tails (1987, 2008, 2020) do not push the closed-form intercept

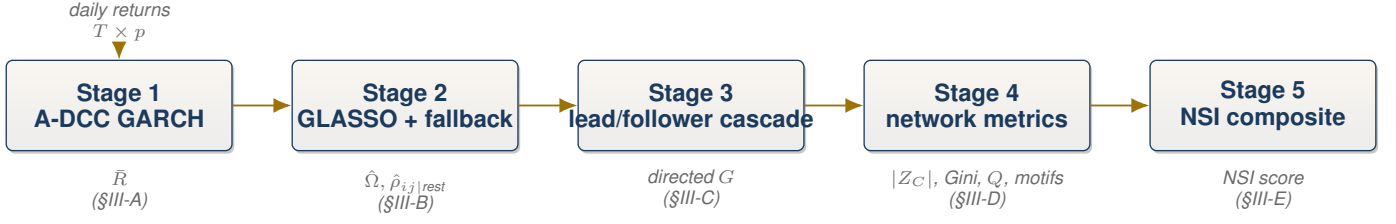


Fig. 1 – Pipeline flow. Each stage’s input arrives from the left (Stage 1 from the raw return panel, Stages 2–5 from the upstream stage), its primary transform is the boxed title, and the artefact passed downstream is annotated below the box. The whole pipeline runs per snapshot; section labels point to the formal treatment of each stage.

$(1 - a - b)\bar{Q} - g\bar{N}$ outside the PSD cone. The optimum is $(\hat{a}, \hat{b}, \hat{g}) = (0.0009, 0.9973, 0.0008)$ with $\hat{a} + \hat{b} + \hat{g} = 0.9990 < 1$. At the optimum $\hat{a}$ and $\hat{g}$ are both $\mathcal{O}(10^{-3})$, so the per-day innovation and asymmetric updates are small and the fitted correlation process is near-integrated; $\bar{R}$ then sits close to the static window sample correlation, with a quantifying ablation folded into F8. Surviving seeds agree within 1.1×10^{-3} , above the 10^{-4} ex-ante threshold, so the multi-modality flag trips; the spread, the $\bar{R}$ insensitivity to the 100-ticker selection, and the deferred CRSP-panel refit are examined in Appendix §C.

The A-DCC quasi-log-likelihood at the optimum is $\mathcal{L}_{\text{DCC}} = 200,718$ over the $100\text{-asset} \times 9,976\text{-day}$ estimation panel.²

For each snapshot (a, b, g) is re-applied within the window on the coverage-filtered sub-panel ($\geq 80\%$ coverage, 655–889 alive) with window-local targets, and the window-average $\bar{R} = T_{\text{win}}^{-1} \sum_{t \in \text{win}} R_t$ feeds the rest of the pipeline. At $n/p < 1$ the rank-deficient target leaves $\bar{R}$ near-PSD with a small negative-eigenvalue leak controlled by n/p ; Stage 2’s shrinkage absorbs it before the GLASSO call (per-snapshot eigenvalues in Appendix §C-F).

B. Stage 2: GLASSO Precision-Matrix Filtering

Sparsifying $\bar{R}^{-1}$ via the Graphical LASSO [7] yields a partial-correlation matrix $\rho_{ij|\text{rest}} = -\omega_{ij} / \sqrt{\omega_{ii}\omega_{jj}}$ and an undirected edge set indexed by $|\omega_{ij}| > 10^{-10}$.

Adaptive EBIC penalty. The penalty λ^* is selected by the Extended Bayesian Information Criterion of Foygel and Drton [16],

$$\text{EBIC}(\lambda) = -2\mathcal{L}(\hat{\Omega}_\lambda) + k \log n + 4k\gamma \log p, \quad (1)$$

on a 25-point log-spaced grid $\lambda \in [0.05, 1.0]$, with γ tiered by n/p . The penalty in eq. (1) reduces to the standard BIC [37] on this panel’s maximum $n/p \approx 0.32$ (the 2024 Contemporary baseline) since none of the snapshots reach the $\gamma > 0$ tiers, whose lower boundary is $n/p = 3$; tier values are reported in Appendix §D. The empty- and sub-degree-graph failure mode observed at $n/p < 0.30$ is addressed by the identity-target shrinkage tiers (Table D.7) and the constrained-BIC fallback (eq. 2) below, not by EBIC itself.

²The strict $\alpha_i + \beta_i < 1$ check fails for 173 of the 1,439 GARCH-fitted tickers (122 above unity at floating-point precision, only 18 with a numerically meaningful excess $> 10^{-6}$, max 1.4×10^{-3} ; 51 at the boundary), admitted because the per-step recursion stays positive at $\alpha_i + \beta_i = 1$ when $\omega_i > 0$ (boundary-exclusion sensitivity deferred; Appendix C).

Identity-target shrinkage and a fallback rule. Across the panel ($p \in [655, 889]$, every window at $n/p < 0.5$) the raw $\bar{R}$ is near-rank-deficient (§III-A) and the sklearn coordinate-descent solver returns “Non SPD result” over most of the grid. We therefore apply Ledoit–Wolf-style identity-target shrinkage $\bar{R} \leftarrow (1 - \delta)\bar{R} + \delta I$ [38] before the GLASSO call, with δ tiered by n/p (Table D.7); the closed-form Ledoit–Wolf optimum is bypassed because short crisis windows do not deliver reliable higher-order moment estimates.

Spectral safety net. A defensive guard $\bar{R} \leftarrow \bar{R} + (10^{-4} - \lambda_{\min}(\bar{R}))I$ for post-shrinkage $\lambda_{\min}(\bar{R}) < 10^{-6}$ never fires on this panel, since the shrinkage tier alone lifts every post-shrinkage $\lambda_{\min}$ above 0.04; post-shrinkage no snapshot loses a grid point, so the only remaining selection question is the degree floor below (Appendix §C-F, §D).

Constrained-BIC fallback for sub-degree graphs. Let $k(\lambda)$ denote the edge count (the number of upper-triangle, $i < j$, non-zero entries) of the GLASSO estimate at penalty λ , and define a structural floor $k_{\min} = \max(p, 10)$: any graph with $k < p$ has average degree below two, which would force the downstream PageRank and Louvain stages to operate on a disconnected or tree-with-deficit input. When the unconstrained BIC argmin lands in the sub-degree region we substitute

$$\lambda_{\text{fallback}}^* = \arg \min_{\lambda : k(\lambda) \geq k_{\min}} \text{BIC}(\lambda), \quad (2)$$

the BIC-optimal λ inside the constrained set, not a heuristic “densest-convergent” rule. The fallback fires on nineteen of the twenty windows (all but the 2024 Contemporary baseline), where the unconstrained argmin lands at the empty graph; the realised λ^* are daggered in Table B.2. The rule is reported as a methodology caveat (Lim. vi, F6 in §IX) rather than folded into a tuned EBIC.

The grid is traversed sparse-to-dense with early stopping, with both argmins defined over the visited subset that covers the BIC minimum on every snapshot; the full Stage-2 routine (EBIC tier, shrinkage, spectral net, and this fallback) is Algorithm 3 of the appendix.

C. Stage 3: Lead/Follower Direction Assignment

Two independent tests, one signed and one unsigned, must both fire for an edge to receive a direction. For each undirected edge (i, j) surviving Stage 2 both tests are run in both directions. The first is a lagged partial correlation (Frisch–Waugh–Lovell, FWL) between the source’s lagged return $r_{s,u-1}$ and

the target's contemporaneous return $r_{t,u}$, controlling for the target's lagged return and for the top-5 contemporaneous assets ranked by $|\bar{R}_{sk}|$; only a positive significant correlation at $\alpha = 0.05$ is admitted as evidence of source-leading-target (a significant negative residual is read as lagged anti-correlation, and Granger continues to supply unsigned evidence below). The second is a bivariate Granger F -test at lag 1 on the same daily horizon.

Two-test cascade. The two tests probe complementary signed/unsigned aspects of the lead/follower structure, and combining them suppresses the single-test false-direction rate (FWL alone fires bidirectionally on 0.1–1.5% of Stage-2 edges, Granger alone on 1.0–8.8%, peak Mar 2020). Each direction d gets a confidence $c_d \in [0, 1]$ (a positive lagged partial correlation sets $c_d \leftarrow \max(c_d, \hat{\rho}_d)$, a significant Granger test $c_d \leftarrow \max(c_d, 0.5)$); a direction wins on $1.5\times$ dominance, else the edge is a mutual dyad, and $c = 0$ edges are dropped. The ad hoc $\alpha = 0.05$ and $1.5\times$ thresholds are addressed in F3. The cascade leaves 3.7–22.3% mutual (Table E.9) and drops 50.3–83.0% of Stage-2 edges; the screening-then-cascade composition is why no panel-wide multiple-testing correction is applied here (Appendix §E-D).

The mutual fraction itself is informative: crisis-mean 13.24% against non-crisis-mean 9.77% (+3.47 pp), with the top of the distribution crisis-dominated even though the regime means are close (full ranking in §VI-C). It is the most directly observable form of the Z_{MR} motif-suppression signal analysed in §V-D.

D. Stage 4: Network Analysis

Each directed network is summarised with four metrics corresponding to four ex-ante hypotheses. Q1 (non-randomness) reports the empirical clustering Z -score of the symmetrised Stage-3 graph against 200 Erdős-Rényi $G(n, m)$ nulls [39] matched in node and edge count, with clustering computed as the mean local clustering coefficient $\frac{1}{p} \sum_i C_i$; $|Z| > 1.96$ rejects randomness at the two-sided 5% level. Q2 (leadership) uses weighted PageRank [22] with damping 0.85 on the directed graph (edges weighted by |partial correlation|), summarised by the Gini of the score distribution and the full-node Herfindahl index $HHI = \sum_i r_i^2$.

Q3 (community structure) runs Louvain [23], [28] weighted-modularity [48] maximisation and reports Q with GICS [50] sector purity $\text{Purity} = (1/p) \sum_c \max_s |c \cap s|$ (11-sector, retroactive; F10). Q4 (recurring patterns) reports three directed MAN triads [25], [26]: FFL (030T), MR (111D), SIM (021D), with Z -scores from 100 rewires of a dyad-preserving null fixing in/out degrees and mutual-pair counts [40], [41], [24]; temporal-motif extensions [42], [27] are deferred since Stage 3 aggregates each window into one snapshot.

E. Stage 5: Network Stress Index

Four channels, weighted by a prior, collapse a directed network into a scalar stress reading on $[0, 1]$. Stage 5 aggregates the Stage-3 directed-edge sparsity, the Stage-4 (§III-D)

PageRank concentration, the Stage-1 mean A-DCC correlation, and the Stage-4 FFL motif shift into a single weighted score:

$$\text{NSI} = w_s \tilde{s} + w_h \tilde{h} + w_\rho \tilde{\rho} + w_\mu \tilde{\mu}, \quad (3)$$

with $(w_s, w_h, w_\rho, w_\mu) = (0.25, 0.20, 0.35, 0.20)$. The four channels are $s = 1 - d$ (sparsity of the Stage-3 directed network with $d = m_{\text{dir}}/\binom{p}{2}$ and mutual pairs counted twice), $h = \sum_{i \in \mathcal{T}_{10}} \tilde{r}_i^2$ (top-10 normalised PageRank Herfindahl, $\tilde{r}_i = r_i / \sum_{j \in \mathcal{T}_{10}} r_j$), $\tilde{\rho}$ (upper-triangle mean of the window-averaged $\bar{R}$), and $\mu = Z_{\text{FFL}} / |Z_{\text{FFL}}^{\text{baseline}}|$, the FFL motif shift against the mean $|Z_{\text{FFL}}|$ across the five calm baseline windows (1993, 2005, 2013, 2017, 2024 Contemporary). Each component is min/max normalised across the twenty snapshots; the composite NSI therefore lies in $[0, 1]$ but only attains 1 when a single snapshot maximises every channel, which does not occur on this panel (sample maximum NSI = 0.758, Oct 2008 GFC).

The weights are a prior, not an estimate: mean correlation carries the strongest systemic signal ($w_\rho = 0.35$), sparsity ($w_s = 0.25$) reflects the prior that GLASSO sparsifies under stress (only partly confirmed, since this channel inherits Stage-2 regularisation), and PageRank Herfindahl ($w_h = 0.20$) and the FFL motif shift ($w_\mu = 0.20$) carry the structural-concentration and triadic channels. The case for refitting them is revisited in §V-E.

IV. DATA AND TEMPORAL SNAPSHOTS

Three construction choices shape every downstream result: the historical-constituent panel, the twenty regime snapshots, and the parallel ADV liquidity panel. The data come from the CRSP daily stock file (CRSP-US Stock, accessed through Bilkent University's WRDS subscription) over 1985-01-02 to 2024-12-31 (10,080 trading days), keyed on point-in-time S&P 500 membership via `crsp.msp500list` and indexed by CRSP permanent identifier (PERMNO). The CBOE volatility benchmark used in Stage 5 is a Whaley [62] VXO/VIX continuity series spliced at 2003-09-22, sourced from FRED (VXOCLS+VIXCLS, the academic-standard public mirror of CBOE's indices).

Unbalanced historical-constituent panel. The CRSP S&P 500 membership list yields 1,443 historical PERMNOs that were index members at least once between 1985 and 2024; PERMNO 48549 (SEDCO INC, delisted 1984-12-24) falls outside the panel start, leaving 1,442 tickers in the active universe. PERMNO indexing, rather than ticker symbols, eliminates the ticker-reuse contamination that affected free-public vendor histories (CFC, SBNY, SW, AMCR), so no symbol-level audit filter is needed. Six 2008-era distress exits (LEH, BSC, MER, FNM, FRE, WAMU) enter Stage 1 with their full pre-bankruptcy trajectory plus CRSP's delisting-aware total return (DLRET) on the bankruptcy date; WAMU, recorded as a trading suspension rather than a formal delisting, carries a null DLRET with the bankruptcy-day return implicit in the final prc trajectory $\$46 \rightarrow \0.16 (Appendix §B-A). Returns enter Stage 1 as $\log(1 + \text{ret})$ on CRSP's split- and dividend-adjusted series. Per-window balanced sub-panels (in-window

coverage $\geq 80\%$) carry 655–889 alive PERMNOs; the always-alive $\geq 99\%$ -coverage subset (278 tickers) supplies the 100-ticker A-DCC fit. Coverage rules, the $N=1,439$ GARCH-fitted analysed panel, and the GICS mapping from CRSP siccd bands are in Appendix §B.

Twenty temporal snapshots (Table B.2): seven crisis peaks, two recoveries, six non-crisis stress windows, and five calm baselines. Event windows (crisis + stress + recovery) are uniform six calendar months peak-centered (≈ 127 trading days); baseline windows are uniform twelve calendar months (≈ 252 days), removing the prior $4.8\times$ ad-hoc window-length variation. Calm baselines are admitted only if the annual VIX $\max < 26$ and $\text{mean} < 16$, which excludes 2011-12 and 2015 (the Aug 2011 EU-debt / US-downgrade and Aug 2015 yuan-devaluation episodes) and Jan 2020 (subsumed by the Mar 2020 + Jun 2020 COVID pair). The average correlation matrices for both modern crises against two baselines are in Fig. C.3.

Snapshot label convention. The labels in Table B.2 name the central month of each regime, not the calendar interval. The two crisis-recovery pairs {Oct 2008 GFC, Mar 2009 Recovery} and {Mar 2020 COVID, Jun 2020 Stable} occupy temporally proximate but non-overlapping windows—the COVID pair contiguous, the GFC pair separated by a 40-day gap—that share no trading days yet sample adjacent macro regimes; the cluster bootstrap of §VII-B treats each pair as a single resampling unit (per-pair calendar gaps in Appendix §J-A).

Two-panel design (coverage vs. liquidity). Coverage selection does not condition on trading volume, so small-cap names with long uninterrupted listings carry the same weight as megacaps. As a built-in robustness check we also construct an ADV panel: the top- N PERMNOs ranked by mean dollar-volume (daily product of CRSP $|prc|$ and vol ; Appendix §B-D). At $N = 500$ on the twenty-snapshot panel the two selection rules overlap at only Jaccard 0.274 because the 40-year point-in-time membership universe carries many long-tenured small-caps with high coverage but modest ADV that the coverage panel keeps and the ADV panel drops. All main-text figures and tables use coverage by convention, and the per-snapshot cross-rule deviations across $N \in \{50, \dots, 500\}$ under both rules are tabulated in Appendix §I (the largest cross-rule $|\Delta NSI|$ on the panel reaches 0.331 at $N = 200$ on the 2022 Rate Hikes snapshot; the headline rank ordering survives). The dual selection is the primary defence against the speculative-tail objection: any reader who suspects low-liquidity tickers drive the result can verify the same numbers on the liquidity-controlled panel.

V. EMPIRICAL RESULTS

Across the twenty snapshots, the seven crisis peaks separate from the thirteen non-crisis windows most sharply on PageRank Gini concentration (the largest panel-wide effect size, Cohen’s $d = 1.04$ [49]) and, among the directed-reciprocity channels, on the mutual-dyad fraction (crisis-mean 13.24% vs. non-crisis-mean 9.77%); the regime-mean direction is preserved on $|Z_{MR}|$ but two long-baseline windows (2024

Contemporary, 2005 Calm) interleave with the crisis tier. Fig. 2 renders the Stage-3 directed networks for the two modern crisis peaks alongside the 2024 baseline; Tables F.10 (Q1/Q2) and F.11 (Q3/Q4) collect the four ex-ante metrics, visualised by regime in Fig. F.7. Density-controlled robustness is in §VI, formal inference in §VII, and consolidated verdicts in §VIII.

A. Q1: Non-randomness

Q1 hypothesises ex-ante that the partial-correlation graph deviates from an Erdős-Rényi null more strongly under crisis. Raw clustering $|Z_C|$ runs from 14.66 to 68.12 and clears the $|Z| > 1.96$ bar in every snapshot, but does not order the regimes: the sample maximum is Sep 2001 9/11 (68.12), yet the 2024 Contemporary (57.0) and 2005 Calm (46.7) baselines sit among the highest while the modern crises are mid-pack (Mar 2020 47.8, Oct 2008 20.7). $|Z_C|$ therefore tracks edge density and window length rather than regime, so Q1 confirms non-randomness on every snapshot but does not discriminate; the substantive discriminators are the density-controlled cross-sector edge fraction (§VI-B) and the mutual-dyad fraction (§VI-C), with the Watts-Strogatz σ alternative unreliable on sparse fallback windows (§VI-A).

B. Q2: Leadership

Aggregate PageRank concentration separates crisis from calm on average: the two highest Gini values are the modern crisis peaks Mar 2020 COVID (0.522) and Oct 2008 GFC (0.454). Crisis-mean Gini 0.403 against non-crisis-mean ($n=13$) 0.333 is a +20.7% increase in the predicted direction, and PageRank Gini is the only channel to clear the formal permutation test ($p = 0.022$, §VII-A). The individual ranks are not a clean calm $<$ stress $<$ crisis ordering, however: the 2024 Contemporary baseline (0.403) and Q4 2018 VolShock stress window (0.406) interleave with the lower crisis peaks, so the metric separates aggregate-crisis from aggregate-calm but not regime-by-regime.

The full-network Herfindahl moves in the same direction at the regime-mean level (crisis-mean $HHI_{full} = 0.00429$ vs. non-crisis-mean 0.00403, a small +0.00026 gap), and the top-10 normalised HHI variant (Table G.15) similarly fails to separate crisis from baseline; the aggregate-leadership claim therefore holds in regime-mean direction on Gini only.

Leader identity does not match a “financial-institution-dominance” reading (top-3 per snapshot in Table F.11): Oct 2008 GFC’s leaders are energy E&P names (CHK, NOV, EP) and Mar 2020 COVID’s are insurance / annuity issuers (MET, LNC, PRU), sector-coherent within each crisis but not reproducing a single crisis-hub identity across crises; the aggregate Gini concentration signal does survive the cross-crisis test.

C. Q3: Community Structure

Raw Louvain Q spans 0.453 to 0.898 and purity 0.812 to 0.982; the seven crisis peaks scatter across almost the full Q range, so raw modularity does not order the panel by regime, and the Q and purity axes decouple (2022 Rate Hikes pairs

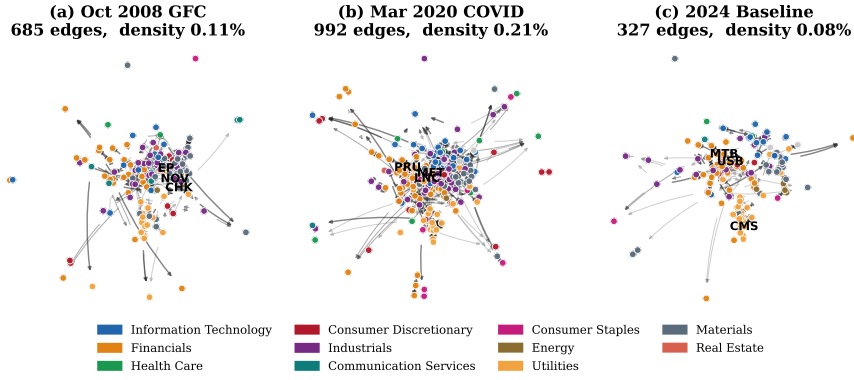


Fig. 2 – Strategic impact maps. (a) Oct 2008 GFC (685 directed edges = 435 one-way + 2×125 mutual). (b) Mar 2020 COVID (992 directed edges = 680 + 2×156). (c) 2024 baseline (327 directed edges = 271 + 2×28). Counts read directly from `stage3_results.pkl`; the in-panel annotations use the same generator (`paper/generate_figures.py`) so caption and figure cannot diverge. Node colours = GICS sector; edge width/opacity binned by $|\rho_{ij}|_{\text{rest}}$; top-3 PageRank tickers per panel labelled. Per-panel density labels use the convenience normalisation $\hat{d} = m_{\text{dir}}/p^2 \approx d/2$; quantitative density references use $d = m_{\text{dir}}/\binom{p}{2}$.

the sample-max Q with non-maximal purity; 2007 Subprime the reverse).

Raw Q is also sensitive to the Stage-2 edge budget (sparse fallback- λ^* graphs give mechanically lower modularity), so the dissolution hypothesis is read from the density-controlled analysis of §VI-B: the cross-sector edge fraction gives crisis-mean 0.679 vs. non-crisis-mean 0.626 ($p = 0.109$), while density-matched modularity at the smallest top-779 budget is inconclusive. Both readings are carried into the verdict (§VIII).

D. Q4: Motif Patterns

Mutual regulation collapses sharply at both crisis peaks; the dyad fraction makes the same signal directly observable. The three MAN triadic motifs counted by Stage 4 are sketched in Fig. F.9. FFL (feed-forward loop, 030T): a source node drives a target both directly and through one intermediate. MR (mutual regulation, 111D): a pair of nodes with edges in both directions plus one outside-pair driver. SIM (single input module, 021D): one node sends edges to two distinct targets without any reciprocal arc.

The motif results support the mutual-regulation-decline hypothesis on aggregate (Fig. F.8). MR (111D) reaches its sample minimum at Mar 2020 COVID ($Z_{\text{MR}} = -23.90$), but 2024 Contemporary and 2005 Calm (both baselines) fill slots 2–3 ahead of Oct 2008 GFC (fourth, -13.05), so $|Z_{\text{MR}}|$ singles out the COVID peak without separating crisis from baseline. On the normalised SP_{MR} the two strongest crises order first (Oct 2008, -0.592) and second (Mar 2020, -0.572).

Feed-Forward Loops trend positively across all regimes (crisis mean $Z_{\text{FFL}} = 14.5$ versus baseline mean 16.6 on the panel) but do not separate regimes; Single Input Modules are uniformly negative across regimes.

The mutual-dyad fraction $f_{\text{mut}} = n_{\text{mutual}}/(n_{\text{directed}} + n_{\text{mutual}})$ is the sharpest of the directed-reciprocity channels (the largest panel-wide standardized effect is PageRank Gini, $d = 1.04$): crisis-mean 13.24% versus non-crisis-mean 9.77% (+3.47 pp, +36% relative), with three of the seven crisis windows in the

top four. It is revisited in §VI-C (full ranking) and §VII-A (permutation test).

E. The NSI and Its VIX Benchmark

Oct 2008 GFC tops the composite at NSI = 0.758, the single clearest crisis signal on the panel. Table G.15 reports the Stage-5 NSI of (3) with its four min/max-normalised component inputs ($1 - d$, HHI, $\bar{\rho}$, μ); Fig. 3 renders it against the Whaley [62] VXO+VIX continuity series (FRED VXOCLS+VIXCLS, splice 2003-09-22).

Crisis vs. non-crisis. The current weights do not separate crisis from the broader non-crisis arm: crisis-mean NSI 0.525 versus non-crisis-mean 0.529 and baseline-mean 0.494, with the stress arm (0.557) and recovery arm (0.532) both above the crisis mean. Oct 2008 GFC ranks first (0.758) but Mar 2020 COVID falls to slot 8 (0.545), and three of the four channels confound regime with density (the sparsity channel inherits Stage-2 regularisation, the FFL motif-shift channel rewards long calm windows, and the top-10 PageRank Herfindahl is too narrowly distributed, range $[0.102, 0.130]$); only the mean-correlation channel cleanly ranks the two modern crises above all baselines. This is the central NSI limitation and motivates the F1 density-invariant-channel redesign of §IX.

(iii) *VIX benchmark:* NSI and the window-mean VXO+VIX continuity correlate at Pearson $r = 0.321$ and Spearman $\rho = 0.212$ (Table G.14, scatter in Fig. G.12); the cluster bootstrap of §VII-B (resampling the GFC and COVID pairs as joint clusters) yields Pearson 95% CI $[-0.289, 0.657]$, Spearman 95% CI $[-0.325, 0.652]$, and a label-shuffle $p = 0.167$.

A rolling NSI counterpart on a simplified pipeline (raw Pearson, fixed $\lambda = 0.15$, no Stage-3 cascade or motif channel; Algorithm 6) gives contemporaneous Pearson $r = 0.203$ against daily VIX. Lagged cross-correlations at 5–63 NSI grid steps are sign-inconsistent (positive at the shortest lags, negative from the ≈ 15 -step lag onward), attributed in Appendix §J to the mean-correlation channel’s VIX mean-reversion. The snapshot-level $r = 0.321$ remains the primary benchmark; the rolling counterpart establishes no leading signal.

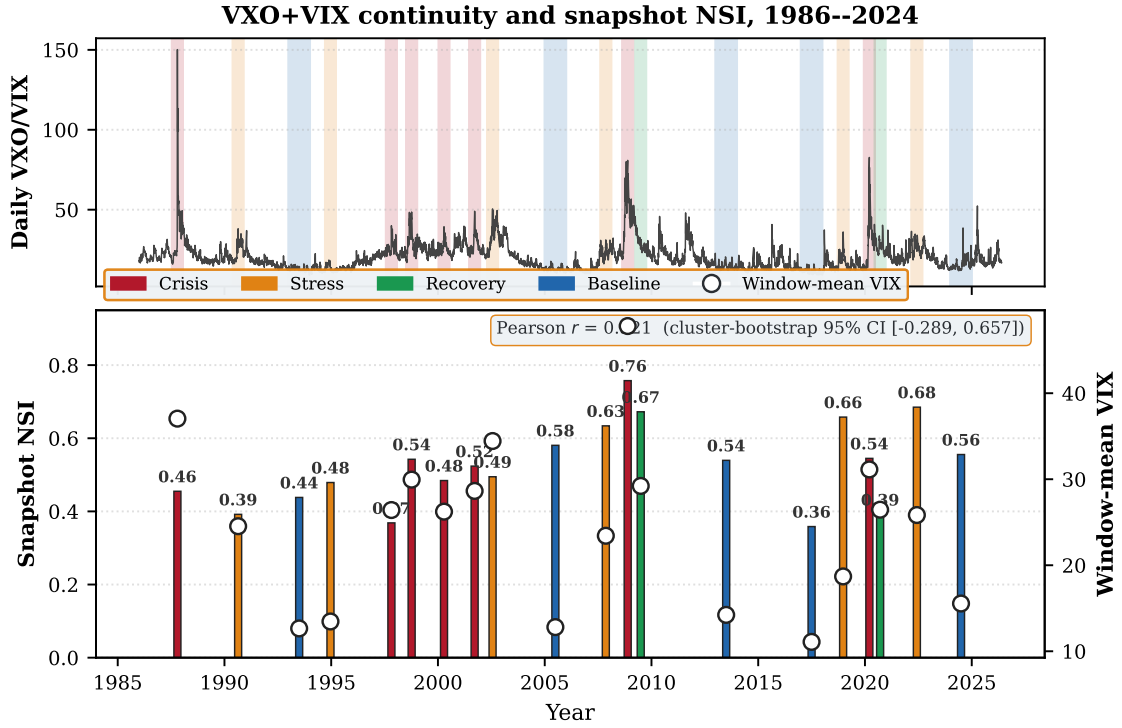


Fig. 3 – NSI vs. VXO+VIX continuity, 1986–2024 (the VXO benchmark begins 1986; the return panel begins 1985). *Top*: daily benchmark (FRED VXOCLS+VIXCLS, Whaley [62] splice 2003-09-22) with snapshot windows shaded by regime; *bottom*: snapshot NSI (bars) and window-mean benchmark (open circles, right axis). Palette ■ crisis, ■ stress, ■ recovery, ■ baseline. Snapshot Pearson $r = 0.321$ (cluster-bootstrap 95% CI $[-0.289, 0.657]$, §VII-B); Oct 2008 GFC takes the top NSI slot but the other six crisis peaks (Mar 2020 at slot 8) interleave with stress and recovery windows under the current calibration (F1, §IX).

A weight-sensitivity sweep (equal, channel-heavy, ± 0.05 , VXO+VIX-fit NNLS, and Dirichlet variants; Appendix §G-D) preserves the Oct 2008 GFC top rank but recovers no sustained crisis-vs-non-crisis ordering on the existing four channels, motivating the F1 channel substitution rather than a weight retune.

Single-snapshot walk-throughs through all five pipeline stages are in Appendix §L: Mar 2020 COVID (the modern-crisis case) and 2017 Calm (the sample-minimum low-NSI counterpoint).

VI. ROBUSTNESS ANALYSIS

Two raw Stage-4 metrics, $|Z_C|$ (Q1) and Louvain Q (Q3), are confounded by the Stage-2 edge density spread; both are re-run with the density confound removed (density-invariant statistics absorb density algebraically; density-matched comparisons threshold each window to a common edge budget; both are documented in Appendix §H). The mutual-dyad fraction is a density-normalised ratio; the PageRank Gini is sparsity-influenced (it rises with the PageRank teleport-floor tail, while the dominant-hub top-10 concentration runs the other way; Appendix §F-B), so a density-matched Gini at a common top- k budget, the analogue of the Q3 cross-sector control, is left as the open test (F4).

A. Small-world coefficient σ for Q1

The Watts-Strogatz-style small-world coefficient $\sigma = (C_{\text{emp}}/C_{\text{null}})/(L_{\text{emp}}/L_{\text{null}})$ [43], [44] normalises clustering by

path length and is density-invariant in principle, with an ex-ante $\sigma > 4$ threshold. On the twenty-snapshot CRSP panel the Stage-3 graphs are too fragmented (the undirected projection splits into 405–718 connected components whose giant component spans only 3–37% of the nodes) for L_{emp} to be well-defined on every component, so σ is withheld and the threshold is recorded as inconclusive (Disclosure 10, Appendix §H-A). The companion clustering metric $|Z_C|$ of Q1 remains the primary clustering-deviation channel.

B. Density-matched community analysis for Q3

Rebuilding each snapshot as a threshold network at a common edge budget removes the edge-budget confound. The threshold network at edge budget k is constructed by ranking all $\binom{p}{2}$ off-diagonal entries of the window-average A-DCC correlation matrix $\bar{R}$ by $|\bar{R}_{ij}|$ magnitude and retaining the top- k as an undirected unweighted graph (Stage-2's selected λ^* does not enter the construction). Two density-control choices on this threshold construction give two different readings.

(a) *Common top- k at the smallest GLASSO graph ($k = 779$).* Crisis-mean Louvain Q (0.603) sits below both the non-crisis ($n=13$) mean (0.693) and the calm-baseline ($n=5$) mean (0.707) in the dissolution-consistent direction but within the wide cross-snapshot spread. This stringent budget censors the dense calm-baseline graphs, so the reading is inconclusive here (crisis-vs-non-crisis $p = 0.092$, $d = -0.66$, set aside on that censoring ground); the Q3 verdict is taken from

the cross-sector fraction (b). Per-snapshot Q and Q_{rel} are in Appendix §H-B.

(b) *Cross-sector edge fraction at a fixed top- k budget.* A single top- $k = 10709$ budget shared across all twenty snapshots (sized so the sparsest panel, 2024 Contemporary at $p_{\text{min}} = 655$, sits at 5% density, falling to $\approx 2.7\%$ at the largest) counts the share of edges crossing GICS sector boundaries (Table H.17). Crisis-mean 0.679 vs. non-crisis-mean 0.626 is higher for crises, as the dissolution hypothesis predicts; the exact-permutation test gives $p = 0.109$, BH $q = 0.191$, so the cross-sector signal corroborates the dissolution direction even though density-matched modularity at the smaller top-779 budget is inconclusive.

The per-snapshot fractions, the wider +12.7% crisis-vs-calm-baseline contrast (0.679 vs. 0.602, with Oct 1987 and 1990-91 at the panel top), and the sector-pair decomposition are in Appendix §H-C.

C. Mutual-dyad fraction for Q_4

The mutual-dyad fraction $f_{\text{mut}} = n_{\text{mutual}} / (n_{\text{directed}} + n_{\text{mutual}})$ measures how much of the surviving temporal structure is reciprocal rather than hierarchical. On the twenty-snapshot panel the top four values are Oct 2008 GFC (22.32%), 2007 Subprime Buildup (19.27%, stress), Mar 2020 COVID (18.66%), and Oct 1987 Black Monday (17.59%); three of the seven crisis peaks hold three of the top four slots, with the 2007 Subprime stress window (the GFC precursor) as the one non-crisis interloper at slot 2. Crisis-mean 13.24% vs. non-crisis-mean 9.77% is a +36% relative gap; baseline-mean 8.06% and stress-mean 10.82% both sit below the crisis arm. This is the strongest crisis discriminator among the directed-reciprocity channels (PageRank Gini concentration carries the largest panel-wide effect size, $d = 1.04$) and is consistent with the MR motif suppression: when more arrows live inside mutual dyads, fewer single outward arrows remain to form MR triads (111D).

D. Cross-Crisis Reproducibility

Running the same pipeline on the GFC (financial-system) and COVID (externally-driven) crises is a built-in robustness test. The two modern crises co-vary across most Stage-4 axes (both top-three on mean $\bar{p}$, SP_{MR} , and the mutual-dyad fraction, both high on PageRank Gini and depressed $|Z_{\text{MR}}|$), and hub identity is sector-coherent within each crisis (energy E&P at Oct 2008, insurance / annuity at Mar 2020) but does not reproduce a single triplet across crises, so the aggregate Gini signal survives the cross-crisis test while hub identity does not. The fourteen-metric GFC-vs-COVID table and the rewired-null caveat are in Appendix §H-E.

VII. STATISTICAL INFERENCE

Three test families convert the descriptive contrasts of §§V–VI into formal verdicts: exact permutation tests of the crisis-vs-non-crisis contrast (§VII-A), cluster- and block-bootstrap confidence sets for the snapshot- and rolling-NSI–VIX correlations (§§VII-B–VII-C), and a multiple-testing correction

for the per-snapshot Z -scores (§VII-D). Independent NumPy default_rng streams seeded at 2026 pin every branch; every value is emitted as a LaTeX macro by paper/_inference.py so the typeset numbers track the cached pipeline.

A. Exact Permutation Tests on Regime Contrasts

Seven crises and thirteen non-crises yield $\binom{20}{7} = 77,520$ unique label assignments, all enumerated exactly; the one-sided exact p -value $p = \#\{\text{perms} : \bar{X}_{\text{crisis}} - \bar{X}_{\text{non-crisis}} \geq \text{observed}\} / 77,520$ has floor 1.3×10^{-5} (power in Appendix §J-G). Benjamini–Hochberg FDR [45] is applied across the seven metrics (Table II); the metrics share Stage-2 graph structure, so the positive-dependence assumption justifying standard BH over Benjamini–Yekutieli [46] is plausible but not formally verified. The within-pair regime dependence of the contiguous COVID pair inflates the permutation null variance and is conservative for the one-sided p ; a COVID-merged sensitivity ($\binom{19}{7} = 50,388$) preserves the rank ordering (Appendix §J-A).

Table II reports the per-channel results: PageRank Gini is the only channel clearing the uncorrected one-sided $p < 0.05$ ($p = 0.022$) and carries the largest effect size ($d = 1.04$), with mean $\bar{p}$ ($d = 0.71$), the mutual-dyad fraction ($d = 0.68$), and cross-sector edges next in descending $|d|$ order; no channel clears Benjamini–Hochberg (smallest $q = 0.152$). The full-network HHI is non-rejecting ($p = 0.293$) but moves in the predicted direction.

TABLE II – Exact permutation tests for the crisis-vs-non-crisis contrast ($n_{\text{crisis}} = 7$, $n_{\text{non-crisis}} = 13$, $\binom{20}{7} = 77,520$ assignments enumerated). One-sided p -value tests crisis $>$ non-crisis; floor $= 1/77,520 \approx 1.3 \times 10^{-5}$. q is the Benjamini–Hochberg FDR-corrected value across the 7-test family. d is Cohen’s d on the same contrast. The cross-sector edge fraction uses a top- $k = 10709$ threshold network shared across snapshots (5% density at the smallest panel $p_{\text{min}} = 655$; auto-sized in `src.stage4_network.crisis_signals`). Every cell is auto-generated from `paper/output/inference_macros.tex`. Rows are ordered by effect size; bold p flags the four largest- $|d|$ channels (a reading aid, not a significance marker). Only PageRank Gini clears the uncorrected one-sided $p < 0.05$, and no channel clears Benjamini–Hochberg.

Metric	Crisis $\bar{X}$	Non-crisis $\bar{X}$	d	p	q
PageRank Gini	0.403	0.333	1.04	0.022	0.152
Mean A-DCC $\bar{p}$	0.345	0.251	0.71	0.076	0.191
Mutual-dyad fraction (%)	13.24	9.77	0.68	0.084	0.191
Cross-sector edges (top- k)	0.679	0.626	0.60	0.109	0.191
PageRank HHI (full-network)	0.0043	0.0040	0.26	0.293	0.374
$ Z_{\text{MR}} $	9.32	8.17	0.22	0.321	0.374
NSI	0.525	0.529	-0.03	0.525	0.525

B. Cluster Bootstrap for the Snapshot NSI–VIX Concordance

The snapshot Pearson $r = 0.321$ (§V-E) treats the twenty snapshots as exchangeable, but the GFC pair {Oct 2008 GFC, Mar 2009 Recovery} and the COVID pair {Mar 2020 COVID, Jun 2020 Stable} occupy temporally proximate windows that share no trading days—the COVID pair contiguous, the GFC pair separated by a 40-day gap—but sample adjacent macro

regimes. Sampling uncertainty is re-estimated under a cluster bootstrap that resamples eighteen clusters with replacement (the two pairs plus the sixteen remaining windows as singletons): $B = 10\,000$ resamples give Pearson 95% CI $[-0.289, 0.657]$ and Spearman 95% CI $[-0.325, 0.652]$, and a label-shuffle ($B = 10^5$) gives permutation $p = 0.167$ (§J-A tabulates the per-pair calendar gaps).

C. Block Bootstrap for the Rolling NSI–VIX Correlation

Three block lengths, half-year to multi-year, bracket the persistence of the rolling concordance. The rolling-NSI contemporaneous Pearson against the VXO+VIX continuity is $r = 0.203$ over $T = 468$ overlapping NSI grid samples (§V-E). A 95% CI is computed under the Politis–Romano stationary block bootstrap [47] with $B = 5\,000$ resamples at three mean block lengths $\bar{L}$ measured in NSI grid steps (each step = 21 trading days ≈ 1 calendar month): $\bar{L} = 6$ (≈ 6 months) gives $[0.012, 0.370]$, $\bar{L} = 12$ (≈ 1 year) gives $[-0.052, 0.398]$, $\bar{L} = 24$ (≈ 2 years) gives $[-0.067, 0.428]$. The CI widens with the block length (widths roughly 0.36–0.50) and the qualitative call is block-length-sensitive: the $\bar{L} = 6$ (six-month) CI marginally excludes zero, while the $\bar{L} = 12$ and $\bar{L} = 24$ CIs include it. Together with the sign-inconsistency of the lagged correlations across horizons (already documented above), the positive contemporaneous correlation does not extend into a robust leading signal at the one- to two-year horizons relevant for early warning. Zero-variance replicates would be dropped from the percentile vector and reported separately; no resample required dropping (5000, 5000, 5000valid resamples at $\bar{L} = 6, 12, 24$).

D. Multiple-Testing-Corrected Per-Snapshot Z-Scores

Every per-snapshot Z -score in Q1 and Q4 is checked against a Bonferroni correction. With twenty snapshots per axis the per-axis Bonferroni-20 two-sided 5% threshold is $|Z| > \Phi^{-1}(1 - 0.025/20) \approx 3.023$; the per-snapshot ranges $|Z_C| \in [14.66, 68.12]$, $|Z_{FFL}| \in [1.18, 27.10]$, $|Z_{MR}| \in [1.21, 23.90]$, and $|Z_{SIM}| \in [0.89, 28.27]$ are read against it per axis. Each axis carries a different ex-ante null, so the correction is within-axis rather than pooled over all $4 \times 20 = 80$ tests; raising the threshold to the pooled Bonferroni-80 value 3.421 leaves the same 7 sparse-end motif cells below threshold and reverses no call, since no $|Z|$ falls in the (3.023, 3.421) band.

VIII. HYPOTHESIS-TEST SUMMARY

No crisis-vs-non-crisis channel survives Benjamini–Hochberg at FDR-5% (smallest $q = 0.152$, PageRank Gini); Gini is the only channel significant at the uncorrected one-sided 5% level ($p = 0.022$), the mutual-dyad, mean- $\bar{p}$, and cross-sector channels following but non-rejecting. Q2 and Q4 hold in regime-mean *direction* on their density-invariant channels, Q3 is partially supported, Q1 is a non-discriminating sanity check, and the NSI composite carries descriptive but not formal support. Verdicts:

(Q1) Sanity check, not a regime discriminator. All twenty snapshots clear the Bonferroni-20 threshold $|Z| > 3.023$

(range 14.66–68.12, §VII-D); the metric tracks edge density and window length rather than regime (§V-A), and the density-invariant small-world coefficient σ is not reported because the Stage-3 graphs are too fragmented for L_{emp} to be well-defined on every connected component (§VI-A).

(Q2) Holds on Gini in regime-mean direction and clears the uncorrected one-sided test; the full-network HHI moves in the predicted direction but is descriptive, not discriminative. Crisis-mean Gini 0.403 vs. baseline-mean ($n=5$) 0.301; the crisis-vs-non-crisis ($n=13$) permutation test uses non-crisis mean 0.333 and yields $p = 0.022$, BH $q = 0.152$. Full-network HHI moves in the predicted direction (crisis-mean 0.0043 vs. non-crisis-mean 0.0040, signed diff 0.0003, Cohen’s $d = 0.26$); the permutation test on HHI returns $p = 0.293$, $q = 0.374$. Hub identity is sector-coherent within each crisis (energy E&P at Oct 2008 GFC, insurance / annuity at Mar 2020 COVID; energy-services at Oct 1997 Asian and Sep 2001 9/11, utilities / financials at Oct 1998 LTCM and Apr 2000 Dot-com, and industrials / materials / media at Oct 1987 Black Monday for the other five) but does not reproduce a single hub triplet across crises.

(Q3) Partially supported. The density-matched cross-sector edge fraction at a top- $k = 10709$ budget (5% density at the sparsest snapshot, the 2024 Contemporary baseline at $p_{\min} = 655$) gives crisis-mean 0.679 vs. baseline-mean ($n=5$) 0.602; the crisis-vs-non-crisis ($n=13$) permutation test uses non-crisis mean 0.626 and yields $p = 0.109$, BH $q = 0.191$. Density-matched modularity at top-779 is inconclusive at this budget, with the smaller budget producing fragmented Stage-3 graphs at most of the seven crisis windows.

(Q4) Holds on the mutual-dyad fraction; $|Z_{MR}|$ holds on the modern crisis pair and is interleaved with long-baseline windows elsewhere. Mar 2020 COVID $Z_{MR} = -23.90$ takes the sample minimum; Oct 2008 GFC (-13.05) sits fourth on $|Z_{MR}|$ with 2024 Contemporary (-14.41 , baseline) and 2005 Calm (-13.17 , baseline) between the two crises on this channel. Crisis-mean $|Z_{MR}| = 9.32$ vs. non-crisis ($n=13$) mean 8.17; crisis-vs-non-crisis ($n=13$) permutation $p = 0.321$, BH $q = 0.374$. The mutual-dyad fraction is the sharpest of the directed-reciprocity channels: crisis-mean 13.24% vs. non-crisis-mean 9.77% gives permutation $p = 0.084$ and Cohen’s $d = 0.68$ (the largest panel-wide effect size is PageRank Gini, $d = 1.04$; full ranking in §VI-C).

(Stage 5 NSI) Not supported on the current weight calibration; motivates F1. Crisis-mean NSI 0.525 sits within ± 0.031 of the baseline (0.494) and non-crisis (0.529) means, with the stress arm (0.557) above the crisis arm; the permutation test gives $p = 0.525$, BH $q = 0.525$, non-rejecting at any conventional FDR threshold. Oct 2008 GFC ranks first (NSI 0.758) but Mar 2020 COVID falls to slot 8 (0.545). Snapshot Pearson against the VXO+VIX continuity is $r = 0.321$ (cluster-bootstrap 95% CI $[-0.289, 0.657]$, label-shuffle $p = 0.167$), positive but carried by Oct 2008 ranking first on both NSI and the benchmark rather than by the composite as a whole. The full per-metric scorecard (BH, BY, COVID-merged, LOO columns) is Table J.27.

IX. LIMITATIONS AND FUTURE WORK

Limitations (i) and (iii)–(x) each carry a paired future-work item (F1 and F3–F10), ordered by impact on the headline NSI claim; limitation (ii) documents resolved survivorship and VIX-benchmark choices and carries no open future-work item (the F2 slot is intentionally vacant).

(i) *NSI crisis-vs-non-crisis separation.* The composite ranks Oct 2008 GFC first but interleaves the remaining six crisis peaks with stress and recovery windows (Mar 2020 COVID at slot 8), and at the regime-mean level the crisis arm sits within ± 0.031 of both the baseline and non-crisis means with the stress arm above both (§V-E). Three of the four channels confound regime with density (sparsity, FFL motif-shift, and the narrowly-distributed top-10 PageRank Herfindahl); the panel-size and selection-rule sensitivity across $N \in \{50, \dots, 500\}$ under coverage and ADV is in Appendix §I and §I-I. **F1:** replace the three density-confounded channels with density-invariant alternatives (cross-sector edge fraction, mutual-dyad fraction, and the SP_{MR} profile that already ranks Oct 2008 GFC and Mar 2020 COVID first and second), and refit the weights via leave-one-crisis-out cross-validation across the seven crisis windows; report the held-out VIX-fit Pearson and crisis-vs-non-crisis effect size.

(ii) *Survivorship and the pre-1990 volatility benchmark.* The panel is survivorship-free by design: keyed on point-in-time S&P 500 membership and indexed by PERMNO, each 2008-era distress exit (LEH, BSC, WAMU, MER, FNM, FRE) enters Stage 1 with its full pre-bankruptcy trajectory and CRSP's delisting-aware total return (DLRET), and PERMNO indexing carries no ticker-reuse contamination (§IV, Appendix §B-A). Two residual caveats remain. (a) Share-class pairs for issuers with multiple PERMNOs (GOOG/GOOGL, FOX/FOXA, NWS/NWSA, DISCA/DISCK; four pairs, eight PERMNOs) currently enter the panel at PERMNO granularity; a PERMCO-level deduplication is deferred to F9, but the empirical impact on the headline metrics is null on this panel (quantified in item (ix) and Appendix §B-E).

(b) Pre-1990 windows (notably Oct 1987) predate the CBOE VIX (begins 1990-01-02), so Stage 5 uses the Whaley [62] VXO+VIX continuity (§IV); figures plot it under the single label “VIX” with the splice disclosed in captions.

(iii) *Lead/follower thresholds ad hoc.* The cascade drops 50.3–83.0% of Stage-2 edges; the $\alpha = 0.05$ and $1.5\times$ thresholds are not joint-sensitivity tested at the headline level. The 14-cell prototype sweep (Appendix §E-E) keeps the crisis-vs-non-crisis f_{mut} mean separation positive (+1.72 to +4.40 pp) on every cell but does not rank every crisis above every non-crisis window. **F3:** extend it to the full 48-cell joint grid over $(\alpha, \tau, \text{lag}, \text{return frequency}, c_{\text{floor}})$, report a permutation- p heatmap for $|Z_{MR}|$ and f_{mut} , and pre-register the headline thresholds on an external registry (e.g. OSF).

(iv) *Density confound, partially resolved.* Top-779 modularity is inconclusive while the top- $k = 10709$ cross-sector fraction is directionally supportive ($p = 0.109$). **F4:** sweep $k \in \{500, \dots, 12000\}$ and report Q , the cross-sector fraction, and the PageRank Gini (a density-matched Gini, testing whether the headline $d = 1.04$ survives a common edge

budget) as functions of k ; compare DC-SBM [59] held-out likelihood against Louvain [23] on the symmetrised graph; extend the 3-type MAN census of §III-D to the full 13-type MAN census [25].

(v) *Diagnostic coverage.* The heatmap covers four of twenty snapshots and impact-map figures cover three of twenty. **F5:** render a 20-panel $\bar{R}$ correlation heatmap grid (one per snapshot, sector-block ordered) and a sector-aggregated leaders panel that collapses the per-snapshot top-3 PageRank tickers into their 11-sector parent for cross-crisis hub-identity comparison; integrate both into the figure pipeline (paper/generate_figures.py) so they regenerate on cache update.

(vi) *Stage-2 fallback rule.* Nineteen $n/p < 0.30$ snapshots invoke the constrained-BIC fallback of eq. (2); the shrinkage tier boundaries and the $k_{\min} = \max(p, 10)$ floor are heuristic, and a panel-scale joint sensitivity over $(\delta, k_{\min})$ is not run on this panel. **F6:** replace the sklearn coordinate-descent solver with the QUIC second-order Newton solver [9]; accompany each λ^* with a debiased-GLASSO confidence set on the partial-correlation entries [60]; sweep the δ tier centres and $k_{\min}$ values and report stability of the headline NSI rank order.

(vii) *Rolling-vs-snapshot NSI scale mismatch.* The rolling NSI uses a simplified pipeline (Algorithm 6: raw Pearson instead of A-DCC, fixed λ , no Stage-3 cascade, no motif channel), so $r = 0.203$ is a lower-bound concordance rather than a degraded snapshot NSI. **F7:** re-run the rolling grid with the full snapshot pipeline, or ablate the snapshot pipeline one channel at a time to attribute the $\Delta r = 0.321 - 0.203$ gap, and verify the contemporaneous Pearson rises into the snapshot cluster-bootstrap CI band.

(viii) *Globally-pooled A-DCC parameters.* (a, b, g) is fit once on the 100-asset balanced subset following [36] but not directly tested per window (Appendix §C-E). **F8:** re-estimate per snapshot (or via block-bootstrap), propagate through Stages 2–5, and compare the asymmetric term g across regimes with block-bootstrap 95% CIs on the per-window estimates, and quantify how far $\bar{R}$ departs from the static window correlation at the fitted near-integrated scalars; a hierarchical (a_w, b_w, g_w) formulation requires Bayesian infrastructure (PyMC/Stan) outside the current frequentist pipeline and is left to follow-up work.

(ix) *Share-class disambiguation deferred.* The CRSP file carries the four share-class pairs (GOOG/GOOGL, FOX/FOXA, NWS/NWSA, DISCA/DISCK) as distinct PERMNOs under a common PERMCO, and the panel of (ii) keeps both PERMNOs of each pair (Appendix §B-E). The network-aggregate inflation is null on this panel: across the 23 pair-snapshot occurrences the Stage-3 cascade emits zero mutual dyads (within-pair $\hat{\rho} \in [0.947, 0.994]$ leaves no lagged structure to score a reciprocal pair; only five lone directed edges form across the 23 occurrences), no pair member reaches the top-10 PageRank list, and under longest-membership-PERMNO collapse full PR Gini drifts $\leq +0.001$, Louvain modularity $|\Delta Q| \leq 0.00639$, and GICS purity $|\Delta P| \leq 0.00501$. **F9:** collapse each pair to the longer-membership PERMNO (a 1,438-PERMNO panel) or retain both with explicit within-pair correlation reporting; both are

documentation-completeness items, not correctness fixes.

(x) *GICS-only sector classification.* Q3 cross-sector and sector-purity metrics use the 11-sector GICS taxonomy; the Fama–French FF-12/FF-49 alternatives [61] are not evaluated, and the 2016 Real-Estate split is applied retroactively. **F10:** re-run under FF-12 and FF-49 alternatives [61] and quantify the bias on the Q3 cross-sector edge fraction and the sector-purity metric, both from the FF substitution and from a static-vs-point-in-time GICS lookup.

X. CONCLUSION

Three of the four ex-ante hypotheses move in the predicted direction on the twenty-snapshot panel, but the NSI composite under its current weight calibration ranks Oct 2008 GFC first while failing to separate crisis from non-crisis at the regime-mean level, motivating the F1 NSI-weight redesign. The integrated pipeline of Contribution (i) runs end-to-end on the 1,442-PERMNO CRSP historical-constituent panel in approximately ~30 min on Apple M5, and the two Stage-2 numerical modifications of Contribution (ii), the n/p -tiered identity-target shrinkage schedule and the constrained-BIC fallback, recover non-empty graphs at every short, high-volatility window (nineteen of twenty fire the fallback) and remain the principal numerical caveat (Lim. vi, F6).

The hypotheses of Contribution (iii) close as follows: Q2 (Gini) is the only channel significant at the uncorrected one-sided 5% level ($p = 0.022$) and carries the largest effect size ($d = 1.04$); Q4 holds in regime-mean direction on the mutual-dyad fraction ($d = 0.68$); Q3 is partially supported on the density-controlled cross-sector edge fraction; and Q1 reduces to a non-randomness check. No channel clears Benjamini–Hochberg FDR-5% (smallest $q = 0.152$). The NSI vs. VXO+VIX concordance of Contribution (iv) gives snapshot Pearson $r = 0.321$ with a cluster-bootstrap CI including zero; Oct 2008 GFC takes the top NSI slot (Table G.14) but the remaining six crisis peaks interleave with stress and recovery windows, the central limitation that motivates F1. The formal inference layer of Contribution (v) and the panel-size sweep in Appendix §I complete the workflow.

a) *Code and data availability.*: Source code at github.com/natozenbilek/StrategicImpactMapping. The 1,442-PERMNO CRSP panel was fetched from the Bilkent University WRDS subscription (2026-05-28); the licence prohibits raw redistribution, so a Zenodo archive (minted on acceptance) carries the derived parquets, the delisting splice trace, the VXO+VIX series (FRED, public), the Stage-1–5 caches, the multipanel summary, and the assertion-suite report (836 checks; 805 PASS, 31 FAIL).

b) *Seeds and determinism.*: All stochastic components use NumPy default_rng streams pinned to seeds 2026 or 42 (per-module assignments in Appendix §M). The Stage-1 multi-start MLE and the Stage-2 sklearn solver are deterministic but non-seeded, with floating-point trajectories depending on the host BLAS; the headline numbers come from a clean-slate Apple M5 / Accelerate run and reproduce within rank order across re-runs, with a formal cross-BLAS comparison left to follow-up work (Appendix §K).

c) *Software stack.*: Python 3.9.6, NumPy 2.0.2 [51], SciPy 1.13.1 [52], pandas 2.2.3 [56], scikit-learn 1.6.1 [53], NetworkX 3.2.1 [54], statsmodels 0.14.6 [55], arch 7.2.0 [57], python-louvain 0.16 [58]. Per-stage complexity, including Stage 2's wall-time dominance, is in Appendix §M-D.

d) *Funding and competing interests.*: No external funding; no competing interests. The CRSP data access is provided by Bilkent University's institutional WRDS subscription (academic-use terms, no equity-trading conflicts), so no human-subjects approval applies.

e) *CRedit authorship contribution statement.*: **Nezih Arhan Tözenbilek** (sole author): all CRedit roles, from conceptualization, methodology, and software through formal analysis, data curation, writing, visualization, and project administration.

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